X. Electron impact on H_2O

The cross sections used for different electron scattering processes on H_2O as a function of energy is shown in Fig. 1. Note that only the total ionization cross-section is shown in the figure. We use in the model those for individual ionization processes. The references for the cross sections are summarized in Table 1.

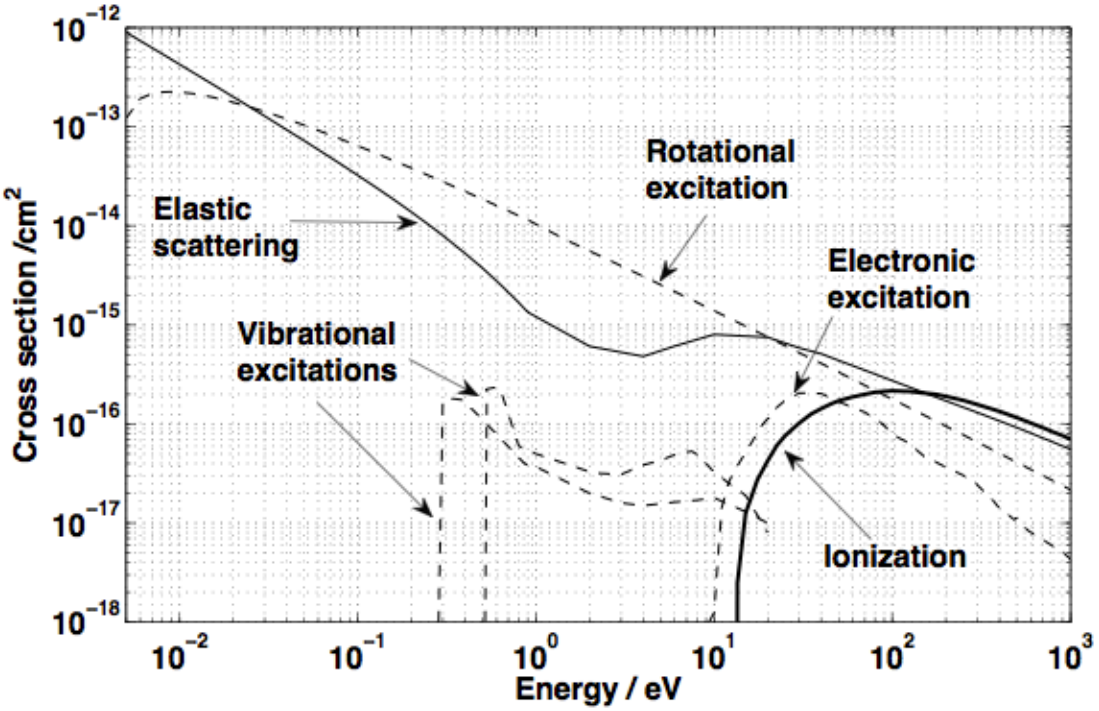


Table 1: References for cross sections (IM05: Itikawa and Mason, 2005; M07: Munoz et al., 2007).

	$E \leq 1$ eV	$1 < E \leq 30$ eV	$E > 30$ eV
Ionization		IM05 (Table 11)	IM05 (Table 11)
Vibrational exciation	IM05 (Table 9)	IM05 (Table 9)	Extrapolation
Rotational excitation	IM05 (J=0-1 in Table 7 and 8)	M07 (Table 1)	M07 (Table 1)
Electronic excitation		M07 (Table 2) combined with IM05 (Table 3) and IM05 (Table 11)	M07 (Table 2) combined with IM05 (Table 3) and IM05 (Table 11)
Elastic scattering	IM05 (J=0-0 in Table 7 and 8)	IM05 (Table 4) combined with M07 (Table 1)	M07 (Table 1)

X.1. Ionization cross section

The electron impact ionization cross sections of H_2O are taken from Table 11 of Itikawa and Mason (2005). Cross sections are given for the production of H_2O^+, OH^+, O^+, O^{++}, H_2^+ and H^+. The last column of the table gives the total ionization cross section, which is equal to the sum of the individual ionization cross-sections and shown in Fig. 1. The energies required for the production of H_2O^+, OH^+, O^+, O^{++}, H_2^+ and H^+ depend on the neutral products as well as the internal energy of the products. We adopt energies from the NIST Chemistry webbook and assign values of 12.62 eV

(for the production of H_2O^+), 18.1 eV (OH^+), 19.0 eV (O^+), 54.1 eV (O^{++}), 20.7 eV (H_2^+) and 16.95 eV (H^+). Note that the energy required for the production of O^{++} has been calculated as the sum of the second ionization energy of O (35.1 eV) and the energy required to produce O^+ from H_2O (19.0 eV).

X.2. Vibrational excitation

The cross sections for the excitation of the symmetric and anti-symmetric stretching modes are combined and taken from Table 9 of Itikawa and Mason (2005). The cross section for excitation of the bending mode is taken from the same table. Cross sections are given for electron energies up to 20 eV in the table. The energy required for the excitation of the stretching modes (bending mode) is approximately 0.453 eV (0.198 eV). Note that we consider here only the lowest vibrational excitations of H_2O molecules starting in their vibrational ground state.

X.3. Rotational excitation

We take the cross sections from Table 1 of Munoz et al. (2007) where energies range from 1 eV to 10 keV. The cross sections were there calculated for an assumed rotational temperature of 300 K. Note, however, that the cross sections at 1, 1.5, 2, 3, 4 and 5 eV are only about 5% higher than the values listed in Table 8 of Itikawa and Mason (2005) for the cross section of the J=0-1 rotational transition. For energies below 1 eV we use the cross sections listed for the J=0-1 rotational transition in Table 7 and 8 of Itikawa and Mason (2005). We assume an avarage energy loss of W_{rot}=4 meV for the electrons in the rotational excitation processes. We later test the sensitivity of the model outputs to the choice of W_{rot}.

X.4. Electronic excitation

The cross-sections for individual electronic excitation processes derived experimentally and theoretically display large (order of magnitude) discrepancies (see Thorn, 2008). Therefore instead of treating the electronic excitation processes individually we focus on the total cross section for electronic excitation.

We use the total scattering cross section, $Q_{total}(E)$, from Table 3 of Itikawa and Mason (2005) and ionization cross sections according to Table 11 of the same work. For energies above 7.5 eV we use the ratios $Q_{EE,ioni}(E)/Q_{total}(E)$ recommended in Table 2 of Munoz et al. (2007) where $Q_{EE,ioni}$ denote the cross section for electronic excitations and ionizations. From the computed $Q_{EE,ioni}(E)$ we then subtract our $Q_{ioni}(E)$ values. This gives us an estimate of $Q_{EE}(E)$ as shown in Fig. 1. For the electronic excitations (at E >10 eV) we adopt an average electron energy loss of W_{EE}=10 eV. We later test the sensitivity of the model outputs to the choice of W_{EE}.

X.5. Pure elastic scattering

The agreement between the total scattering-cross sections derived theoretically and experimentally by Munoz et al. (2007) improves with increasing energy. We select to use the pure elastic cross sections calculated by Munoz et al. (2007) for E > 30 eV. For 1< E ≤ 30 eV we use Table 4 in Itikawa and Mason (2005), which gives cross sections for vibrationally elastic scattering and from these values we subtract the cross sections that we use for rotational excitation. For E<1 eV we use the cross sections

provided for J=0→0 in Table 7 and Table 8 of Itikawa and Mason (2005). The shape (and magnitude) of the pure elastic cross section is rather similar to the one for the momentum-transfer cross section for electron impact on H_2O (see Fig. 7 of Itikawa and Mason, 2005). Note that if we would set the elastic cross section at a constant value of 2×10^{-15} cm^2 for E<1 eV it would have a negligible effect on the output of the model calculations.